

Predictive Modeling of Forest Biomass for Carbon Management

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I am grateful to Jerzy Wieczorek for his guidance and support.
All errors are my own.

I. Introduction

As a data analyst for the United States Forest Service (USFS), one of my responsibilities is to support forest monitoring and land management decisions using data-driven tools. In recent years, tracking forest carbon has become increasingly important because forests play a major role in storing carbon and reducing the impacts of climate change. Accurate estimates of where forest biomass exists help the USFS identify areas that contribute most to carbon storage, guide conservation efforts, and allocate resources for field surveys more efficiently. Since sending field crews to directly measure trees is expensive and time-consuming, developing a predictive model can help the agency estimate carbon presence in places where on-the-ground measurements are not available.

Dataset Description

The dataset used in this study comes from the Forest Inventory and Analysis (FIA) program, which combines field measurements with environmental data collected across the western United States. Each observation represents a 90×90 meter land pixel, and the dataset includes several thousand sampled pixels from four ecological sections in the western U.S. The response variable is `CARBON`, which measures the amount of carbon stored in the aboveground portion of trees. For this project, the response variable is converted into a binary classification problem: pixels are labeled as **Presence** if carbon is greater than 0, and **Absence** if carbon equals 0. This creates a classification task rather than a regression task.

The explanatory variables are all auxiliary variables that can be collected remotely without sending crews into the field. These include satellite-based vegetation indices such as `ndvi`, `evi`, and `tcc16`, which capture greenness and canopy cover; topographic variables such as `elev`, `tri`, `tpi`, `eastness`, and `northness`, which describe terrain; and climate variables such as precipitation (`ppt`) and temperature measures (`tmean`, `tmax`, and related metrics). The dataset also includes a categorical land cover variable, `tnt`, which classifies areas as tree or non-tree based on satellite imagery. Because these predictors are available at large scales and relatively low cost, they are ideal for building a predictive system that can supplement field-based carbon inventories.

Objectives and Methods

The goal of this study is to build a strong **classification model** that predicts whether a given land pixel contains carbon-storing forest biomass. A reliable model would allow the USFS to prioritize field visits to areas where carbon is likely present, while reducing unnecessary surveys in areas unlikely to contain forest biomass. This can save both time and funding while improving the agency's ability to monitor large and remote landscapes. In addition, these predictions could be useful for other organizations, such as state forestry agencies or environmental researchers, who need updated biomass maps for wildfire planning, conservation, or carbon accounting. To evaluate different approaches, I will apply three distinct statistical methods that represent fundamentally different modeling strategies:

1. **LASSO-Penalized Logistic Regression:** A parametric approach that addresses the extreme multicollinearity within the climate data by applying a penalty to shrink redundant predictors to zero.
2. **Support Vector Machine (SVM):** A kernel-based method that maps auxiliary data into a higher-dimensional space to identify an optimal separating hyperplane, allowing for the capture of non-linear boundaries between ecological classes.
3. **Random Forest:** An ensemble method that utilizes a multitude of de-correlated decision trees to capture complex interactions and non-linearities while remaining robust to the identified class imbalance.

II. Exploratory Descriptive Analyses (EDA)

All exploratory data analysis was performed using only the training partition to avoid data leakage and preserve the independence of the test set. This ensures that the test data serves as an unbiased measure of model performance on unseen

observations. Any use of test data during exploratory analysis or model development could artificially inflate predictive accuracy and compromise the validity of the final evaluation.

The training data reveals a substantial class imbalance in the response variable, reflecting the ecological characteristics of the study region. The majority class is **Presence**, accounting for 83.6% of the observations ($n = 3584$), while the minority class, **Absence**, accounts for only 16.4% ($n = 702$). This indicates that most sampled locations in the western United States contain measurable aboveground carbon, while relatively few correspond to zero-carbon areas.

Given this imbalance, a simple majority-class classifier provides an important baseline for comparison. By always predicting **Presence**, the baseline model achieves an accuracy of 83.6%, equal to the proportion of the dominant class. This demonstrates that overall accuracy alone is not an adequate performance metric, since a model can appear highly accurate without learning meaningful relationships from the predictor variables.

For this reason, model performance must also be evaluated using **sensitivity** and **specificity**. Sensitivity measures the model's ability to correctly identify locations with carbon presence, while specificity measures its ability to correctly identify locations without carbon. In the context of the US Forest Service, both metrics are critical: high sensitivity ensures that carbon-rich forest areas are not overlooked, while improved specificity indicates that the model can distinguish true absence rather than simply reproducing the dominant class pattern.

Table 1: Class distribution of the response variable in the training data

Class	Proportion	Count
Presence	0.836	3584
Absence	0.164	702

Vegetation indices derived from satellite imagery appear to be the strongest drivers of carbon presence in the dataset. The exploratory boxplots for `tcc16`, `ndvi`, and `evi` show a clear separation between the two classes, indicating that vegetation-related variables carry substantial predictive information. In general, locations in the **Presence** class tend to have higher canopy cover and stronger vegetation signals, which is consistent with expectations for denser forested areas.

Among the variables, `tcc16` (Analytical Tree Canopy Cover) provides the most distinct separation. Most **Absence** pixels fall below roughly 20% canopy cover, suggesting that low canopy density is strongly associated with non-forest or zero-carbon areas. The vegetation indices `ndvi` and `evi` also differ between classes, but with more overlap. They also show a few extreme low-value outliers within the Presence class. This likely reflects real-world variability in remote sensing data, where vegetation signals can be weaker in arid regions, seasonal landscapes, or high-elevation areas even when some carbon is still present.

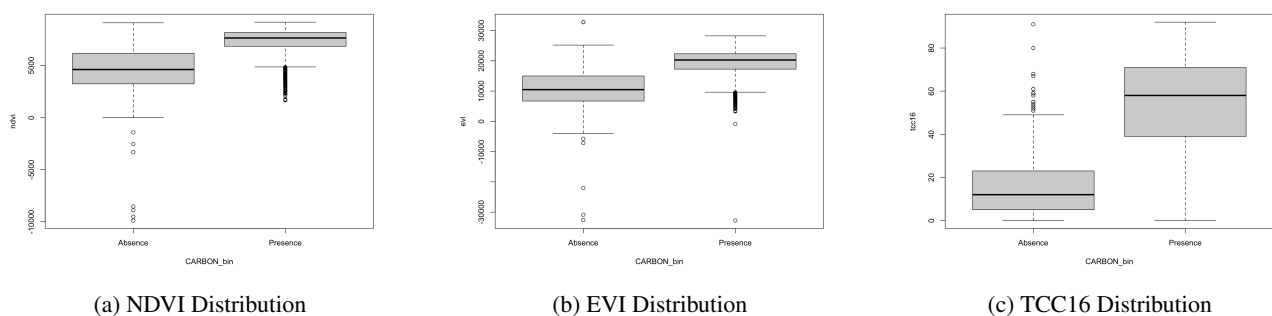


Figure 1: Distribution of vegetation indices and canopy cover by carbon presence class.

The correlation matrix reveals strong multicollinearity within the PRISM climate variables. In particular, a tightly connected "climate cluster" emerges among `tmean`, `tmax`, `tmin`, `tmin01`, `vpdmax`, and `tdmean`, all of which are highly correlated ($r > 0.90$). Including all of these variables in a standard logistic regression model would likely lead to unstable coefficient estimates and make it difficult to separate the individual effects of each climate factor.

A similar pattern is observed in the vegetation indices, where `ndvi` and `evi` are almost perfectly correlated, suggest-

ing that they carry largely redundant information about vegetation greenness. Together, these issues highlight the need for a modeling approach that can handle redundancy in the predictor set.

For this reason, we use **Lasso regression**, which is well-suited for situations like this because it automatically reduces redundancy by shrinking less informative variables toward zero, while retaining only the most useful predictors within highly correlated groups.

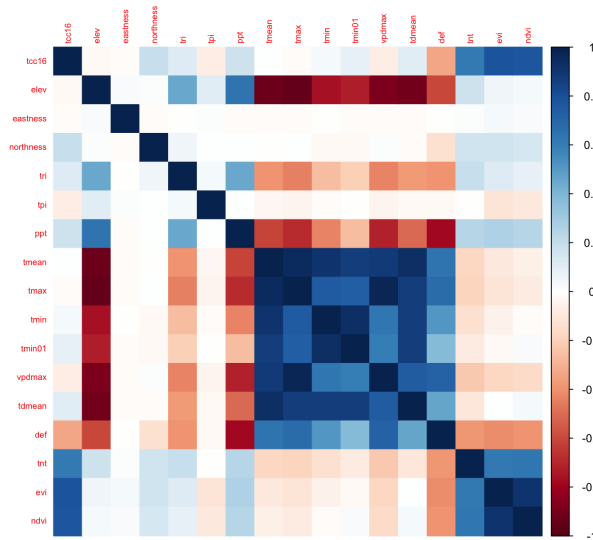


Figure 2: Correlation matrix of numeric predictor variables in the training data

The exploratory plots suggest that the relationships between predictors and carbon presence are likely non-linear. In particular, carbon presence appears to follow a threshold-like pattern: once canopy cover or vegetation indices reach a certain level, the probability of carbon presence increases rapidly before eventually leveling off. This indicates that simple linear assumptions may not fully capture the underlying ecological structure. In addition, interactions are likely to exist between climate conditions (`ppt`) and topography (`elev`), since water availability and elevation jointly constrain vegetation growth and, ultimately, carbon storage capacity in high-elevation or moisture-limited regions.

Based on these characteristics, the three selected models were chosen to capture complementary aspects of the data. The **Lasso regression** addresses multicollinearity among the climate variables by shrinking redundant predictors toward zero and retaining a simpler, more interpretable subset of variables. The **Support Vector Machine (SVM)** with a radial basis function (RBF) kernel is well suited for capturing non-linear decision boundaries and handling outliers by implicitly mapping the predictors into a higher-dimensional space. Finally, the **Random Forest** is expected to deliver the strongest predictive performance because it naturally captures complex non-linear relationships and interactions between variables, while also being robust to class imbalance in the response.

III. Classification Procedures and Results

This section presents the modeling framework used to predict carbon presence and compares the performance of three machine learning approaches: LASSO-penalized logistic regression, Support Vector Machines (SVM), and Random Forests. All models were trained on the training partition ($n = 4,286$) and evaluated on an independent hold-out test set ($n = 4,286$). Performance is assessed using accuracy, sensitivity (ability to detect Presence), and specificity (ability to detect Absence).

Method 1: LASSO-Penalized Logistic Regression

We begin with a LASSO (Least Absolute Shrinkage and Selection Operator) logistic regression. This method is well suited to the dataset because it handles multicollinearity while also performing automatic variable selection through

regularization.

Model Estimation and Selection: The model was fitted using 10-fold cross-validation via the `glmnet` package. The penalty parameter λ was selected using the 1-standard-error (1SE) rule, resulting in $\lambda \approx 0.018$. This choice favors a more parsimonious model, reducing the risk of overfitting while maintaining strong predictive performance.

Feature Selection Results: The LASSO procedure significantly reduced model complexity by shrinking 16 of the 21 predictors to zero. This leaves a compact set of 5 variables, suggesting that carbon presence is primarily driven by a small number of key environmental signals rather than the full predictor set.

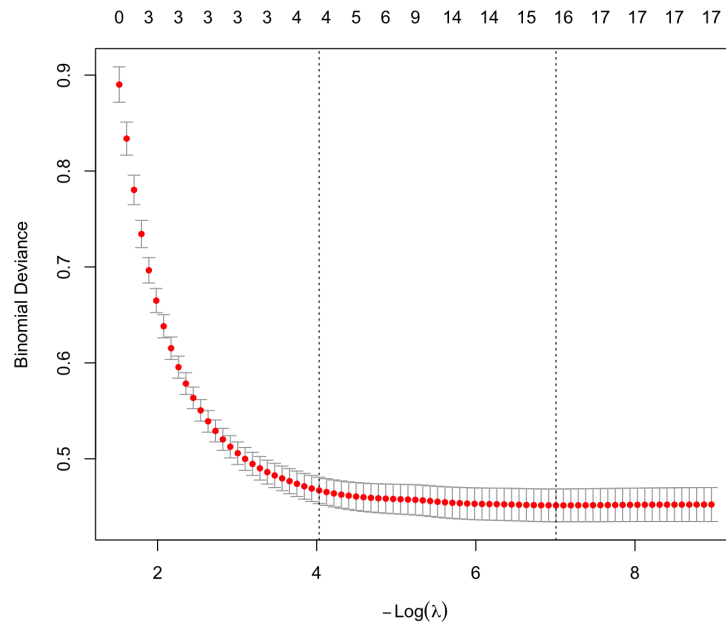


Figure 3: LASSO cross-validation paths; the dashed lines indicate λ_{min} and λ_{1se} .

Final Model Interpretation: The retained predictors and their estimated coefficients ($\hat{\beta}$) are:

- **tnt (0.7577):** strongest positive predictor, highlighting the importance of land cover classification.
- **tcc16 (0.0720):** canopy cover, confirming vegetation density as a key signal.
- **tri (0.0081):** terrain ruggedness, associated with slightly higher carbon presence in more varied landscapes.
- **ndvi (0.0001):** weak but retained vegetation signal.
- **tmax (-0.00002):** slight negative association with maximum temperature.

Method 2: Support Vector Machine (SVM)

To capture potential nonlinear relationships that may not be well represented by a linear model, we implement a Support Vector Machine (SVM) with a **Radial Basis Function (RBF)** kernel. This approach is particularly useful in settings where class separation depends on complex, nonlinear boundaries in the predictor space.

Hyperparameter Tuning: The model was tuned using a grid search with 10-fold cross-validation to identify the optimal combination of parameters. The best-performing configuration was:

$$\text{Cost} = 10, \quad \gamma = 0.01$$

The relatively high cost suggests a stronger penalty for misclassification, leading to a tighter fit around the training data, while the low value of γ ensures that the influence of individual support vectors remains localized and smooth, preventing

the model from becoming overly sensitive to noise and allowing for better generalization across the high-dimensional feature space. .

Model Performance: On the test set, the SVM achieved:

- Accuracy: **90.8%**
- Sensitivity: **97.0%**

Overall, these results indicate that the SVM is highly effective at identifying carbon-positive locations, with particularly strong performance in minimizing false negatives, which is important for ensuring that areas of ecological importance are not overlooked.

Method 3: Random Forest

We next implement a Random Forest model consisting of 500 decision trees. This ensemble approach improves predictive performance by aggregating across many de-correlated trees, allowing the model to capture complex nonlinear relationships and interactions between predictors without requiring strong parametric assumptions.

Variable Importance: Variable importance is assessed using the Mean Decrease in Gini index (Figure 4). Consistent with earlier results, `tcc16` emerges as the most important predictor, followed by `evi` and `ndvi`. This consistency across models reinforces the importance of vegetation structure as the primary driver of carbon presence.

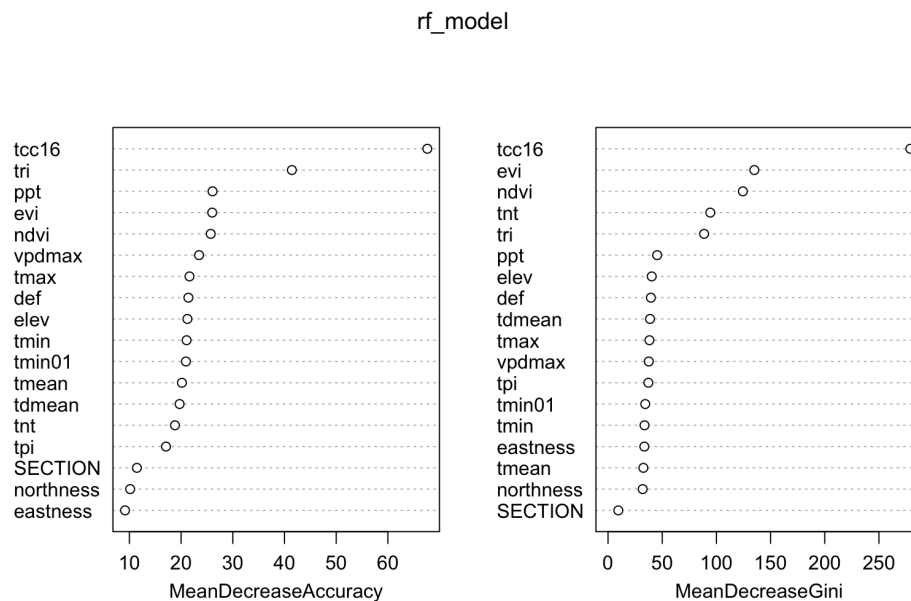


Figure 4: Random Forest variable importance (Mean Decrease in Gini).

Model Performance: The Random Forest achieves a test accuracy of **90.8%**, matching the SVM. However, it offers improved flexibility in capturing nonlinear interactions and is generally more robust to noise and complex feature relationships in the data.

Comparison of Model Performance

To contextualize these results, all models are compared against a baseline classifier that always predicts the majority class (Presence), which yields an accuracy of 83.6%.

Table 2: Classification performance comparison across all models.

Metric	Baseline	LASSO (1SE)	SVM (RBF)	Random Forest
Test Accuracy	83.6%	88.4%	90.8%	90.8%
Sensitivity	100%	95.8%	97.0%	97.0%
Specificity	0.0%	51.1%	60.1%	60.1%

Overall, all three machine learning models significantly outperform the baseline classifier, confirming that meaningful predictive structure exists in the environmental variables. The LASSO model provides strong interpretability through a reduced five-variable structure, while the SVM and Random Forest achieve the highest predictive performance. Among these, the Random Forest offers the most balanced option for practical use due to its combination of strong accuracy, robustness, and interpretable variable importance measures.

IV. Summary and Conclusion

Overall, the results demonstrate that predicting carbon presence using remotely sensed environmental variables is both feasible and practically useful. All three machine learning models significantly outperform the baseline classifier (83.6%), confirming that meaningful ecological signal exists in the predictor set rather than simple class imbalance alone driving performance.

Among the three approaches, the Support Vector Machine (SVM) and Random Forest achieve the strongest and identical predictive accuracy of 90.8%, followed by the LASSO model at 88.4%. From an applied perspective, this level of performance is sufficient for US Forest Service use cases, particularly for large-scale screening, prioritization of field surveys, and regional monitoring rather than exact pixel-level classification.

A key consideration in this context is sensitivity, since failing to identify true carbon-positive areas is more costly than over-prediction. Both the SVM and Random Forest achieve high sensitivity (97.0%), making them reliable first-pass tools for identifying regions that warrant further field validation. However, this strong sensitivity comes with a tradeoff, as all models exhibit lower specificity, indicating a tendency to incorrectly classify some low-vegetation or non-forest areas as carbon presence.

These differences in performance are closely tied to the structure of the models themselves. The LASSO model prioritizes interpretability by reducing 21 predictors down to a small subset of 5 key variables (`tcc16`, `tnt`, `tri`, `ndvi`, and `tmax`). This result is scientifically meaningful, as it highlights vegetation density and land cover as the dominant drivers of carbon presence, with smaller contributions from terrain and climate. In contrast, the Random Forest achieves the best balance of accuracy and flexibility, as it can capture nonlinear relationships and interactions that linear models cannot represent. This makes it more effective at distinguishing true absence from sparsely vegetated regions, where simpler models are more prone to error.

Despite strong overall performance, several limitations should be acknowledged. First, the lower specificity across all models suggests a consistent tendency toward over-predicting carbon presence, likely due to spectral confusion in remotely sensed vegetation indices such as NDVI and EVI, which may respond to grasses, shrubs, or seasonal greenness rather than true forest biomass. Second, strong multicollinearity among climate variables reduces interpretability and introduces redundancy, even though regularization in the LASSO model partially mitigates this issue. Finally, limitations in satellite resolution and temporal alignment between environmental datasets introduce unavoidable noise that constrains predictive accuracy.

From a US Forest Service perspective, these results highlight the practical value of machine learning as a decision-support tool for forest monitoring and carbon accounting. In particular, the Random Forest model provides the most balanced solution, combining strong predictive performance with interpretable variable importance measures. Operationally, these models can be used to prioritize field validation in high-probability carbon areas, reduce monitoring costs by supplementing FIA plot sampling with spatial predictions, and support national-scale carbon assessments in remote or inaccessible regions.

Importantly, these models are not intended to replace field-based forestry measurements, but rather to complement

them by providing scalable, data-driven screening tools that extend the reach and efficiency of existing monitoring systems. In conclusion, the results demonstrate that machine learning models can effectively capture spatial patterns of carbon presence, and that integrating statistical learning with forest inventory data offers a powerful framework for improving large-scale environmental monitoring and management.

A. Technical Supplement: Model Outputs and Diagnostics

This appendix provides the comprehensive statistical outputs and diagnostic visualizations referenced in the main report.

A.1 Exploratory Data Analysis (EDA) Summaries

To establish the statistical baseline and verify data distributions, a full summary of the training dataset was generated.

```
> summary(train_df)
SECTION          tcc16          elev          eastness
Length:4286      Min.   : 0.00      Min.   : 390      Min.   : -100.000
Class :character  1st Qu.:29.00      1st Qu.: 930      1st Qu.: -74.750
Mode  :character  Median :51.00      Median :1208      Median :  -3.000
                        Mean  :48.28      Mean  :1259      Mean   :  -2.417
                        3rd Qu.:69.00      3rd Qu.:1552      3rd Qu.: 70.000
                        Max.   :92.00      Max.   :2616      Max.   : 100.000

northness        tri          tpi          ppt
Min.   : -100.000  Min.   : 0.125    Min.   : -89.37500  Min.   : 256.0
1st Qu.: -70.000  1st Qu.: 10.875   1st Qu.: -4.00000   1st Qu.: 580.2
Median :  -3.000  Median : 19.250   Median : 0.00000    Median : 825.0
Mean   :  -2.439  Mean   : 20.803   Mean   : 0.07277    Mean   : 889.2
3rd Qu.: 65.000  3rd Qu.: 29.500   3rd Qu.: 4.37500    3rd Qu.:1145.0
Max.   : 100.000  Max.   :143.875   Max.   : 78.50000    Max.   :2604.0

tmean          tmax          tmin          tmin01          vpdmax
Min.   : -145.0    Min.   : 335      Min.   : -625.000   Min.   : -1238.0   Min.   : 447.0
1st Qu.: 431.0     1st Qu.: 970      1st Qu.: -132.000   1st Qu.: -893.0    1st Qu.: 865.2
Median : 585.0     Median :1164      Median : 14.000     Median : -726.0    Median :1060.0
Mean   : 561.8     Mean   :1133      Mean   : -9.107     Mean   : -744.8     Mean   :1033.0
3rd Qu.: 712.8     3rd Qu.:1328      3rd Qu.: 122.000    3rd Qu.: -601.0    3rd Qu.:1208.8
Max.   :1073.0     Max.   :1688      Max.   : 459.000    Max.   : -283.0     Max.   :1632.0

tdmean          def          tnt          evi          ndvi
Min.   : -785.00   Min.   : 40.0     Min.   :0.00000     Min.   : -32768    Min.   : -9890
1st Qu.: -195.00   1st Qu.:270.0     1st Qu.:1.00000     1st Qu.: 15056     1st Qu.: 6171
Median :  -47.00   Median :326.0     Median :1.00000     Median : 19512     Median : 7453
Mean   :  -81.25   Mean   :340.4     Mean   :0.8136      Mean   : 17979     Mean   : 6884
3rd Qu.: 65.00    3rd Qu.:393.0     3rd Qu.:1.00000     3rd Qu.: 22018     3rd Qu.: 8095
Max.   : 308.00    Max.   :789.0     Max.   :1.00000     Max.   : 32767     Max.   : 9162

CARBON_bin
Absence : 702
Presence:3584
```

Figure 5: Full Summary Statistics: Distribution of environmental and topographic predictors ($n = 4,286$).

Table 3: Class distribution of the response variable in the training data

Class	Proportion	Count
Presence	0.836	3584
Absence	0.164	702

The following boxplots (Figure 6) illustrate the distributional separation of vegetation signals between areas of carbon presence and absence.

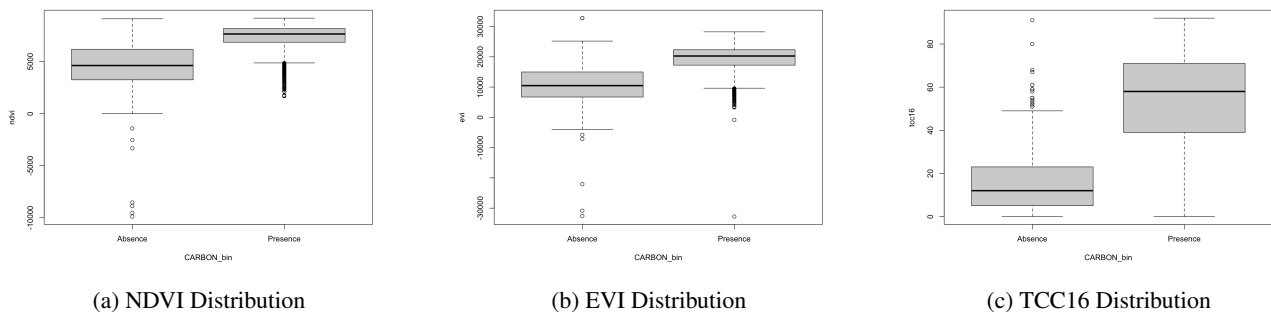


Figure 6: Distribution of vegetation indices and canopy cover by carbon presence class.

A.2 Model 1: LASSO Diagnostics

The LASSO model identified the most parsimonious feature set. The plot below illustrates the cross-validation error across varying levels of λ .

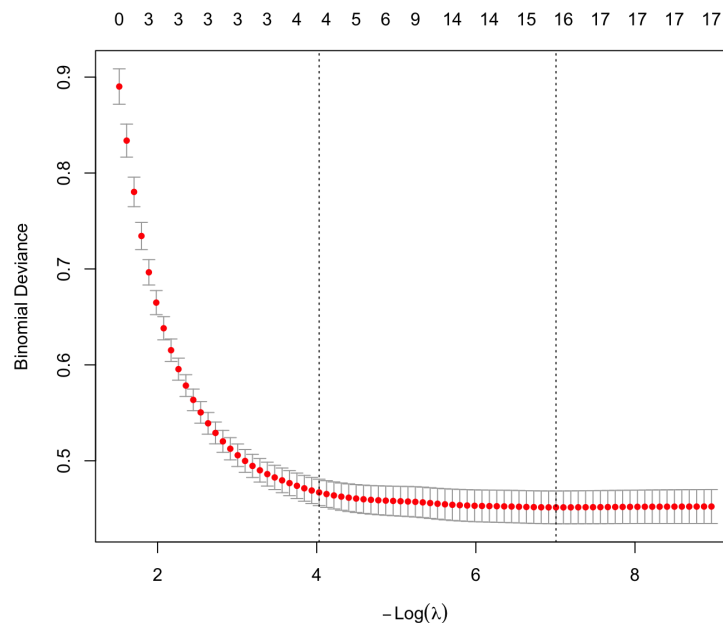


Figure 7: LASSO Selection Path: The 1SE threshold (right-most dashed line) was used to ensure model simplicity.

A.3 Model 2: Support Vector Machine (SVM)

The SVM was optimized using a grid search over cost and gamma parameters. The final results (Table 4) highlight the high sensitivity achieved by the radial basis function kernel.

Table 4: SVM Hyperparameter Results

Parameter	Optimal Value
Kernel	Radial Basis Function (RBF)
Cost (C)	10
Gamma (γ)	0.01
Test Sensitivity	97.0%

A.4 Model 3: Random Forest Importance

The Random Forest provides a more granular look at variable influence. Figure 8 ranks variables by their contribution to overall model accuracy.

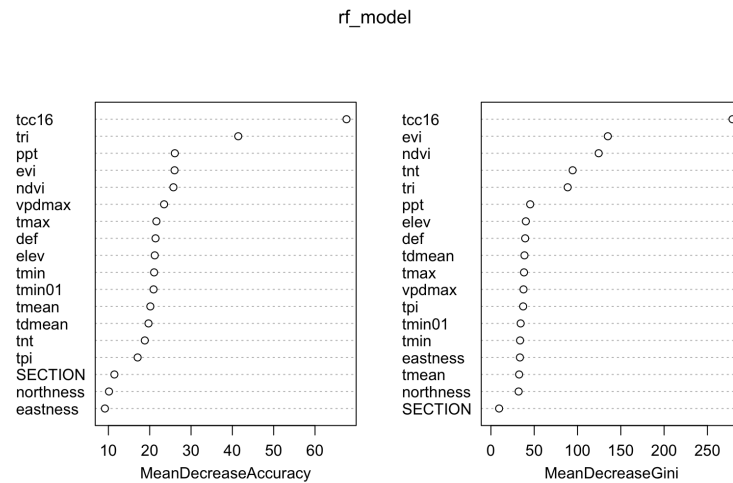


Figure 8: Variable Importance Plot: Mean Decrease in Accuracy and Mean Decrease in Gini.

A.5 Final Performance Metrics

The following screenshot provides the detailed confusion matrix and statistical indices for the best-performing ensemble model.

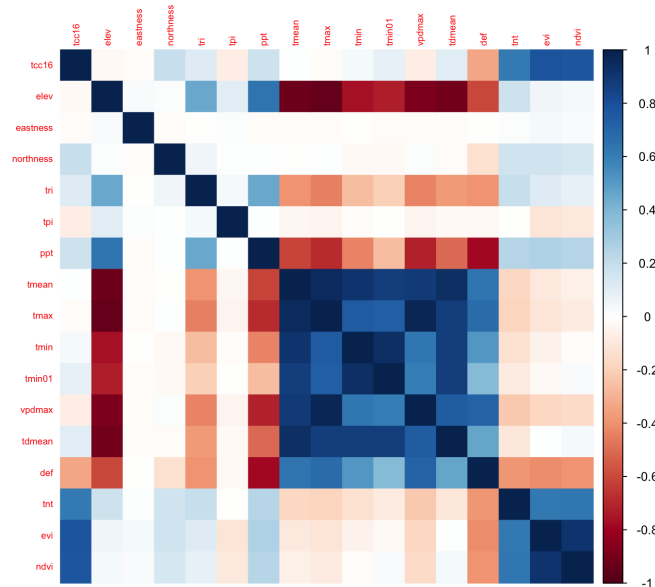


Figure 9: Comprehensive performance statistics and confusion matrix for the test dataset.